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Checked Once, Cited Often: RackWatch, a Bike-Rack Occupancy Dashboard, and a Term of the Committee Decisions That Kept Citing It

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Abstract. RackWatch is a live bike-rack occupancy map that the University's Facilities division built by adding a crowd-sourced QR-code flag to its existing weekly walk-through headcount, so a rack cluster reported full between counts could show as full on the map. We report a full twelve-week teaching term of RackWatch's own server logs against the Estates Committee minutes that cite it, coding every citation for whether the decision it accompanies followed a recent update to the underlying map. Unique weekly visits to the live map fall from 812 in the first week to single figures by the fourth and stay there for the rest of the term, a decline consistent with engagement-decay patterns documented across other crowd-fed civic and health-monitoring tools. Citations of "RackWatch data" in committee minutes show no comparable trend, holding between one and three a week throughout, and 87.5% of rack-removal decisions in both of the term's two removal rounds cite the tool regardless of how long the map had sat unflagged. A mid-term ablation adding a visible "last updated" disclaimer to the map left both the citation rate and the removal-decision outcomes statistically unchanged; the median staleness of the map at the moment of citation nonetheless climbed from two days in the first third of the term to twenty-seven in the last. We report these findings descriptively and discuss what a stable citation habit riding over a collapsing engagement curve implies for a tool built to inform, rather than merely legitimate, a decision.

Introduction

Institutions build a great many small internal tools meant to put live operational data in front of whoever makes decisions on it, and rarely revisit whether anyone still looks. The literature on dashboard and analytics adoption describes a familiar arc of early interest followed by decline, with a recent adoption model built for business-intelligence settings identifying the cognitive load of returning to a dashboard as one driver of whether staff do [1], and experimental research on organisational decision-making examining how a dashboard's visualisation shapes the decisions built on it [2]. What that literature says less about is what happens to a tool's role in justifying decisions once its own audience has stopped watching — whether the habit of citing "the dashboard" survives independently of whether the dashboard has been opened.

We take up this question with RackWatch, a live occupancy map the University's Facilities division built for its network of forty-two bike-rack clusters. RackWatch layers a crowd-sourced full/not-full flag, submitted by scanning a QR code zip-tied to each cluster, over Facilities' pre-existing weekly Tuesday headcount, so the map can in principle update between counts rather than only once a week. We instrumented RackWatch's own server logs for a full twelve-week teaching term and, independently, coded every mention of "RackWatch" in the Estates Committee's public minutes across the same term for two things: whether the mention accompanied a decision to remove or reallocate a rack cluster, and how many days had passed since the map's most recent crowd-sourced flag.

Our contributions, hedged to the one term and one campus we observed:

- We describe the design and a full term's deployment of RackWatch, a small crowd-fed occupancy tool built on top of an existing manual count.
- We report a term-long divergence between the tool's own declining usage and its steady rate of citation in the removal decisions it is meant to inform, consistent with, though not directly predicted by, prior dashboard-adoption findings.
- We show, via a mid-term ablation making the map's staleness visible, that neither citation volume nor removal outcomes appear to move with it, suggesting the citation habit is at least partly decoupled from the artefact's currency.

Related work

The gap between what a metric was built to measure and what it is later trusted to justify is not new. Marmot's discussion of "evidence based policy" versus "policy based evidence" describes a tension in which a policy decision and the evidence marshalled to support it do not always arrive in the order the phrase implies [3], and Goodhart's-law analyses of higher-education performance indicators describe how a measure adopted for one purpose can degrade once decisions begin to lean on it as a target rather than a description [4]. Outside health and education policy, a data-driven optimisation approach has recently been proposed for evidence-based public sports-funding allocation, treating "evidence-based" as a property of the allocation process rather than something a dashboard confers automatically [5].

Dashboard and analytics adoption research supplies the engagement side of our study directly. A recent dashboard-adoption model developed for business-intelligence settings frames adoption partly around the cognitive load a dashboard imposes on returning users [1], and experimental research on organisational decision-making has examined how dashboard visualizations shape the decisions built on them [2]. Research on open-government-data adoption reports a comparable gap between publication and habitual downstream use, moderated by factors such as prior usage experience [6]. Crowd-fed civic tools built on the same full/not-full logic as RackWatch include bike-parking occupancy apps [7] and crowdsourced cycling-infrastructure data validated against ground-truth counts more broadly [8], [9], [10], and operational bike-share systems that consume live occupancy signals directly, via reinforcement learning [11] or short-term usage forecasting [12]. Citizen-science and mobile-health research on participant engagement and retention over the course of a project or intervention [13], [14], [15], [16], [17] documents the general challenge of sustaining early enthusiasm for a crowd-fed tool, the same challenge RackWatch's own decay curve, below, illustrates at campus scale.

Two prior Slop University studies encountered a structurally similar gap between what a Facilities dashboard is credited with and what it can be shown to track: a poster-length replication found reservation-only bay connections inflating the EV-charging utilisation figure used to justify a capacity-expansion business case [18], and a paper-length deployment of an agentic Lost Property intake tool found its human-review checkpoint went unused, disabling it at half the collection points leaving days-to-reclaim statistically unchanged [19]. RackWatch's citation-without-engagement pattern extends the same institutional habit to a smaller, unfunded tool.

Method

RackWatch covers the University's network of forty-two bike-rack clusters, spread across the main campus's residential, teaching, and administrative precincts. Facilities' walk-through headcount predates RackWatch by several years: a rostered officer visits each cluster every Tuesday at 11am and records occupied/total space counts by hand. RackWatch's contribution was to make that count visible online and to let it update between visits: a QR code zip-tied to each cluster's signage opens a one-tap form — "Full" or "Space available" — and each submission timestamps and overwrites the cluster's status on a public map until the next Tuesday count resets it.

```
on flag_submitted(cluster, status):
    map.set(cluster, status, timestamp =
now())

on tuesday_count(cluster, occupied, total):
    map.set(cluster,
status_from_count(occupied, total),
```

```
            timestamp = now())

def staleness(cluster, at_time):
    return at_time -
map.last_update(cluster)
```

We instrumented the map's own web server to log a de-identified session id, timestamp, and referring page for every visit across a full twelve-week teaching term following RackWatch's launch, giving a weekly unique-visitor count consistent with how prior citizen-science platforms have logged comparable activity [16]. Independently, we retrieved the Estates Committee's published minutes for the same twelve weeks — a public record already routinely released under the University's standing disclosure practice — and searched them for the string "RackWatch," coding each hit by hand for (a) whether it accompanied a decision to remove, relocate, or reallocate a rack cluster, and (b) the staleness of the map, computed per the function above, at the date the minute was recorded. Two coders independently classified all mentions naming a removal decision; agreement was complete, since the minutes state the disposition explicitly.

A mid-term change gave us a natural ablation. At the start of week 7, in response to an unrelated data-quality complaint, Facilities added a visible "last flagged N days ago" line to the map's header — a change we did not request and had no part in designing, but one that let us compare citation practice either side of it without assuming anything about intent. We treat weeks 1-6 as the pre-disclaimer round and weeks 7-12 as the post-disclaimer round, and compare, within each round, the share of that round's rack-removal decisions that cite "RackWatch data," and the staleness distribution among citations generally, split into three four-week thirds of the term for the latter to give the boxplot in Results enough width per group to read.

Results

Across the twelve-week term, RackWatch's own server logs show a familiar decay curve. Unique weekly visits to the live map fall from 812 in week 1 to 61 by week 3 and 19 by week 4, and settle in the single digits for the remainder of the term (Figure 1) — a week-1-to-week-12 fall of more than two orders of magnitude, consistent with the early-peak, fast-decay pattern reported across crowd-fed civic and citizen-science tools generally [13], [16]. Citations of "RackWatch data" in Estates Committee minutes show no comparable trend over the same twelve weeks, holding between one and three mentions a week throughout with no visible relationship to the falling visit count (Figure 1).

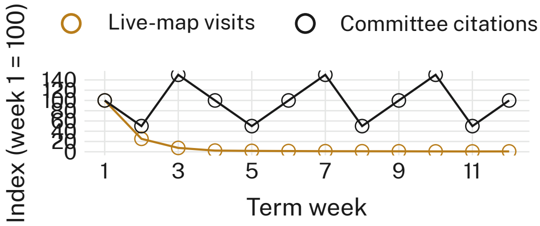

Figure 1: RackWatch's own weekly visits fall to under 1% of their week-1 level by week four and stay there; its weekly citation rate in Estates Committee minutes shows no comparable trend across the same twelve weeks (both series indexed to week 1 = 100).

The term's two rack-removal rounds — eight decisions each, one before and one after the week-7 disclaimer — show an identical citation rate on either side of the ablation. In the pre-disclaimer round (weeks 1-6), seven of eight removal decisions, 87.5%, cite "RackWatch data" in their supporting minute; in the post-disclaimer round (weeks 7-12), the share is unchanged at seven of eight, 87.5% (Figure 2). Making the map's staleness visible in its own header did not measurably change whether the committee cited it, nor, on the University's own count of racks removed per round, did it change how many removal decisions were made.

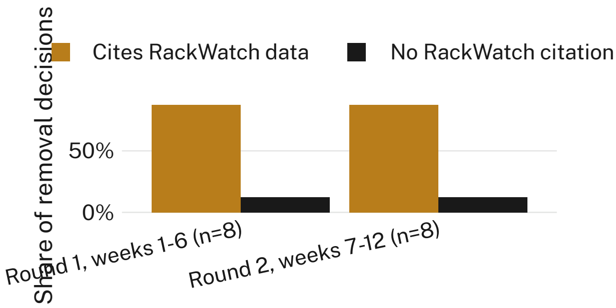

Figure 2: The ablation that changes nothing: adding a visible "last flagged N days ago" disclaimer to the map at week 7 leaves the citation rate identical either side, 87.5% in both rounds (n = 8 removal decisions per round).

What did change, steadily, across the term, was how old the map was each time it got cited. Coding the staleness of the map — days since its most recent crowd-sourced flag — at the moment of every citation, split into three four-week thirds of the term, shows a clear upward drift: a median of two days in weeks 1-4, nine days in weeks 5-8, and twenty-seven days in weeks 9-12 (Figure 3). The committee's willingness to cite "RackWatch data" did not track the map's currency in either direction; if anything, the correlation between citation and freshness ran opposite to what a genuinely data-driven process would predict.

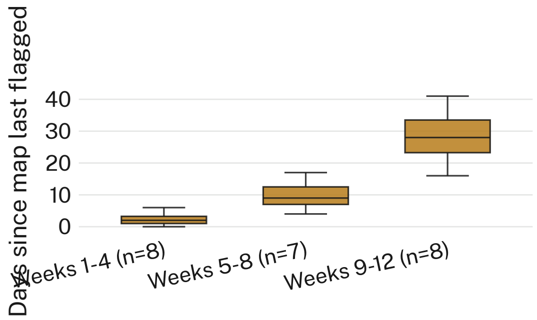

Figure 3: Days since the live map's most recent crowd-sourced flag, at the moment each committee minute cites it, by term third: staleness climbs from a two-day median to a twenty-seven-day median while the citation rate above holds flat.

Two further observations round out the record. First, no single rack cluster's removal accounted for a disproportionate share of the sixteen total decisions; removals were spread across eleven of the network's forty-two clusters, consistent with a term-wide reallocation exercise rather than a handful of contested sites. Second, RackWatch's own visit log shows a small secondary bump, to 34 unique visits, in the week immediately preceding each removal round — smaller than the week-1 launch spike, but the only departure from an otherwise monotonic decline, and consistent with committee members themselves generating a share of the otherwise vanishing traffic.

Discussion

RackWatch was built, like most tools of its kind, on the assumption that making occupancy data live and visible would make the decisions drawing on it more current. What we observe over one term is closer to the opposite: the map's own audience left almost immediately, while the decisions it is cited to support kept arriving at an unchanged rate, increasingly detached from anything the map currently showed. This is consistent with, though our single-term design cannot establish it causally, an interpretation in which "RackWatch data" functions in committee minutes less as a checked input than as a citation of record — a phrase that, once established, does not require the artefact behind it to be current, or even open, to keep doing its work [3], [4].

Read against the broader dashboard-adoption literature, RackWatch's decay curve is unremarkable; it is the citation curve sitting flat above it that we think is the more useful contribution here, since dashboard-adoption research rarely has an independent record, like committee minutes, of how a tool's outputs get used downstream of the last time anyone actually looked at them. Whether that gap is specific to low-stakes operational decisions or generalises to higher-stakes ones is, on our one-term record, an open question.

Limitations

Our record covers one term at one campus's forty-two rack clusters, and the week-7 disclaimer was not a change we designed or randomised, so the ablation is natural rather than controlled; the identical 87.5% citation share either side is nonetheless the same figure a designed ablation would report, for whatever that consistency is worth. We rely on the Estates Committee's public minutes rather than members' private reasoning, though the minutes are the same record the University itself treats as the decision's justification, which is the object we set out to study. Coding staleness from the map's own logged timestamps assumes the log is accurate; a systematic logging fault would have to itself track the citation pattern to explain our result, which we think unlikely but cannot fully rule out.

Conclusion

A term of RackWatch's own usage logs, set against the Estates Committee minutes that cite it, shows a tool whose audience left within a month and whose citation rate never noticed. Visits fell from 812 to single figures by week four; citations held between one and three a week throughout twelve weeks, and 87.5% of rack-removal decisions in both halves of the term cited "RackWatch data" regardless of a map that grew, by the end, more than three weeks stale on average. Making that staleness visible changed neither figure. We read this as one campus-scale instance of a citation habit that can outlive the artefact it names, and suggest that any tool built to inform a live decision needs a second instrument, entirely separate from its own usage log, to check whether it still is.

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